

The empirical recurrence for OEIS A299652: recovering a conjecture that is not written down

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Abstract

OEIS A299652 records “Empirical recurrence of order 89”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 139$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 89, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 89 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A299652 is “Number of $n \times 6$ 0..1 arrays with every element equal to 0, 1, 2, 3, 4, 5 or 6 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

32, 2048, 116654, 6856789, 401989538, 23582064542, 1383316377321, 81146123707386, ...

The entry states:

Empirical recurrence of order 89 (see link above)

As of the “Last modified” line on the live entry (Feb 15 2018 14:41:35, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4096$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 139$ of the 4096, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 139$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 278$ exact terms of the model returns order 89 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{89} c_i a(n-i),$$

c_1	45	c_{24}	−160241288129692710	c_{47}	−37074847921775609017664	c_{70}	14860655
c_2	685	c_{25}	703597124967186412	c_{48}	37574858190677741649486	c_{71}	1739698
c_3	6896	c_{26}	2358379438036629620	c_{49}	98415317451825207321024	c_{72}	3723158
c_4	7388	c_{27}	748126339532418538	c_{50}	59712939215567419834835	c_{73}	−1403658
c_5	−587642	c_{28}	−12005769150289406372	c_{51}	−17610836782760742044873	c_{74}	1748841
c_6	−5137785	c_{29}	−24709124643214372990	c_{52}	−88898327540238688236011	c_{75}	−409274
c_7	−22488936	c_{30}	8174126914712582666	c_{53}	−68788681658031817262905	c_{76}	959397
c_8	121035777	c_{31}	128452456556491090997	c_{54}	−16146290869081949184463	c_{77}	−210504
c_9	1388353946	c_{32}	172670027085128805342	c_{55}	46626562124286795518096	c_{78}	337895
c_{10}	4641703813	c_{33}	−160786530127593364449	c_{56}	50375215779377890146301	c_{79}	−46248
c_{11}	−18962560598	c_{34}	−911167696715097227186	c_{57}	31755716379107646206876	c_{80}	152189
c_{12}	−180728750867	c_{35}	−827758896029194702936	c_{58}	−7255797950091347413407	c_{81}	−26640
c_{13}	−358448598872	c_{36}	1249252657625433882401	c_{59}	−19737116223750379998995	c_{82}	−13054
c_{14}	2113145698980	c_{37}	4453682835085866219314	c_{60}	−20988934504762543670006	c_{83}	−16355
c_{15}	12806201225242	c_{38}	2864852340451596432962	c_{61}	−5709878614657499013897	c_{84}	10403
c_{16}	9666729162413	c_{39}	−5710915084408300163693	c_{62}	2438886660983004204831	c_{85}	16722
c_{17}	−143151426513450	c_{40}	−15503464004664200419602	c_{63}	5923382807754865173092	c_{86}	7005
c_{18}	−522113640252669	c_{41}	−7819565736280809835220	c_{64}	3736715786220558800007	c_{87}	−617
c_{19}	223134548673155	c_{42}	16812398458851664741486	c_{65}	480309484321399849164	c_{88}	−183
c_{20}	5826877662001538	c_{43}	39384229407458282457264	c_{66}	−352528524902534518402	c_{89}	153
c_{21}	12378923897166837	c_{44}	18332524477142358567420	c_{67}	−978952633528360932019		
c_{22}	−22847934809531269	c_{45}	−32280502610597189398486	c_{68}	−119033511438587420340		
c_{23}	−146873940773101905	c_{46}	−73463367865884694523636	c_{69}	−149411965596135500156		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 89, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $89 + 4$ terms removed returns order 89 as well, so $\deg q_{\min} = 89 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $89 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 89 that this sequence satisfies — and that family turns out to have exactly one member.

References

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